

Table 7.1 Experimental estimation-error variance (50 time steps) for Example 7.1 when there is a cross covariance M between the process noise and the measurement noise. The standard filter assumes that $M = 0$, and the correlated filter uses the correct value of M

Correlation M	Standard Filter ($M = 0$ assumed)	Correlated Filter (correct M used)
0	0.076	0.076
0.25	0.030	0.019
-0.25	0.117	0.052

7.2 COLORED PROCESS AND MEASUREMENT NOISE

Our derivation of the Kalman filter in Chapter 5 assumed that the process noise and measurement noise were both white. In this section, we will show how to deal with colored process noise, and we will present two methods for dealing with colored measurement noise.

7.2.1 Colored process noise

If the process noise is colored, then it is straightforward to modify the system equations and obtain an equivalent but higher-order system with white process noise [Buc68]. Then the standard Kalman filter equations can be applied. For example, suppose that we have an LTI system given as

$$x_k = Fx_{k-1} + w_{k-1} \quad (7.16)$$

where the covariance of w_k is equal to Q_k . Further suppose that the process noise is the output of a dynamic system:

$$w_k = \psi w_{k-1} + \zeta_{k-1} \quad (7.17)$$

where ζ_{k-1} is zero-mean white noise that is uncorrelated with w_{k-1} . In this case, we can see that the covariance between w_k and w_{k-1} is equal to

$$\begin{aligned} E(w_k w_{k-1}^T) &= E(\psi w_{k-1} w_{k-1}^T + \zeta_{k-1} w_{k-1}^T) \\ &= \psi Q_{k-1} + 0 \end{aligned} \quad (7.18)$$

The 0 arises because w_{k-1} is independent from ζ_{k-1} , and ζ_{k-1} is zero-mean. We see that w_k is colored process noise (because it is correlated with itself at other time steps). We can combine Equations (7.16) and (7.17) to obtain

$$\begin{aligned} \begin{bmatrix} x_k \\ w_k \end{bmatrix} &= \begin{bmatrix} F & I \\ 0 & \psi \end{bmatrix} \begin{bmatrix} x_{k-1} \\ w_{k-1} \end{bmatrix} + \begin{bmatrix} 0 \\ \zeta_{k-1} \end{bmatrix} \\ x'_k &= F' x'_{k-1} + w'_{k-1} \end{aligned} \quad (7.19)$$

This is an augmented system with a new state x' , a new system matrix F' , and a new process noise vector w' whose covariance is given as follows:

$$\begin{aligned}
 E(w'_k w'^T_k) &= \begin{bmatrix} 0 & 0 \\ 0 & E(\zeta_k \zeta_k^T) \end{bmatrix} \\
 &= Q'_k
 \end{aligned} \tag{7.20}$$

Now the standard Kalman filter can be run on this augmented system that has white process noise, as long as we know $E(\zeta_k \zeta_k^T)$. Computational effort increases because the state vector dimension has doubled, but conceptually this is a straightforward approach to dealing with colored process noise.

7.2.2 Colored measurement noise: State augmentation

Now suppose that we have colored measurement noise. Our system and measurement equations are given as

$$\begin{aligned}
 x_k &= F_{k-1}x_{k-1} + w_{k-1} \\
 y_k &= H_k x_k + v_k \\
 v_k &= \psi_{k-1}v_{k-1} + \zeta_{k-1} \\
 w_k &\sim N(0, Q_k) \\
 \zeta_k &\sim N(0, Q_{\zeta k}) \\
 E[w_k w_j^T] &= Q_k \delta_{k-j} \\
 E[\zeta_k \zeta_j^T] &= Q_{\zeta k} \delta_{k-j} \\
 E[w_k \zeta_j^T] &= 0
 \end{aligned} \tag{7.21}$$

The measurement noise is itself the output of a linear system. The covariance of the measurement noise is given as

$$\begin{aligned}
 E[v_k v_{k-1}^T] &= E[(\psi_{k-1}v_{k-1} + \zeta_{k-1})v_{k-1}^T] \\
 &= \psi_{k-1}E[v_{k-1}v_{k-1}^T]
 \end{aligned} \tag{7.22}$$

There are a couple of ways to solve the colored measurement-noise problem. It was solved by Richard Bucy for continuous-time problems in [Buc68]. Here we will solve the discrete-time problem by augmenting the state. This was originally proposed in [Bry65] in the context of continuous-time systems. We augment the original system model as follows:

$$\begin{aligned}
 \begin{bmatrix} x_k \\ v_k \end{bmatrix} &= \begin{bmatrix} F_{k-1} & 0 \\ 0 & \psi_{k-1} \end{bmatrix} \begin{bmatrix} x_{k-1} \\ v_{k-1} \end{bmatrix} + \begin{bmatrix} w_{k-1} \\ \zeta_{k-1} \end{bmatrix} \\
 y_k &= [H_k \quad I] \begin{bmatrix} x_k \\ v_k \end{bmatrix} + 0
 \end{aligned} \tag{7.23}$$

This can be written as

$$\begin{aligned}
 x'_k &= F'_{k-1}x'_{k-1} + w'_{k-1} \\
 y_k &= H'_k x'_k + v'_k
 \end{aligned} \tag{7.24}$$

This system is equivalent to the original system but has a modified state x' , state transition matrix F' , process noise w' , measurement matrix H' , and measurement

noise v' . The covariance of the process noise and the covariance of the measurement noise are computed as

$$\begin{aligned} E[w'_k w'^T_k] &= E \left[\begin{pmatrix} w_k \\ \zeta_k \end{pmatrix} \begin{pmatrix} w_k^T & \zeta_k^T \end{pmatrix} \right] \\ &= \begin{bmatrix} Q_k & 0 \\ 0 & Q_{\zeta_k} \end{bmatrix} \\ E[v'_k v'^T_k] &= 0 \end{aligned} \quad (7.25)$$

We see that there is no measurement noise, which is equivalent to saying that the measurement noise is white with a mean of zero and a covariance of zero. Theoretically, it is fine to have zero measurement noise in the Kalman filter. In fact, Kalman's original paper [Kal60] was written without any restrictions on the singularity of the measurement-noise covariance. But practically speaking, a singular measurement-noise covariance often results in numerical problems [May79, p. 249], [Ste94, p. 365]. For that reason we will present another approach to dealing with colored measurement noise in the next section.

7.2.3 Colored measurement noise: Measurement differencing

In this section we present a method for dealing with colored measurement noise that does not rely on augmenting the state vector. This approach is due to [Bry68]. As in the previous section, our system is given as

$$\begin{aligned} x_k &= F_{k-1}x_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ v_k &= \psi_{k-1}v_{k-1} + \zeta_{k-1} \\ w_k &\sim (0, Q_k) \\ \zeta_k &\sim (0, Q_{\zeta_k}) \\ E[w_k w_j^T] &= Q_k \delta_{k-j} \\ E[\zeta_k \zeta_j^T] &= Q_{\zeta_k} \delta_{k-j} \\ E[w_k \zeta_j^T] &= 0 \end{aligned} \quad (7.26)$$

Now we define an auxiliary signal y'_k as follows:

$$y'_{k-1} = y_k - \psi_{k-1}y_{k-1} \quad (7.27)$$

Substitute for y_k and y_{k-1} in the above definition of y'_{k-1} to obtain

$$\begin{aligned} y'_{k-1} &= (H_k x_k + v_k) - \psi_{k-1}(H_{k-1}x_{k-1} + v_{k-1}) \\ &= H_k(F_{k-1}x_{k-1} + w_{k-1}) + v_k - \psi_{k-1}(H_{k-1}x_{k-1} + v_{k-1}) \\ &= (H_k F_{k-1} - \psi_{k-1}H_{k-1})x_{k-1} + H_k w_{k-1} + v_k - \psi_{k-1}v_{k-1} \\ &= (H_k F_{k-1} - \psi_{k-1}H_{k-1})x_{k-1} + (H_k w_{k-1} + \zeta_{k-1}) \\ &= H'_{k-1}x_{k-1} + v'_{k-1} \end{aligned} \quad (7.28)$$

H'_{k-1} and v'_{k-1} are defined by the above equation. We see that we have a new measurement equation for the measurement y'_{k-1} that has a measurement matrix

H'_{k-1} and measurement noise v'_{k-1} . Our new but equivalent system can therefore be written as

$$\begin{aligned} x_k &= F_{k-1}x_{k-1} + w_{k-1} \\ y'_k &= H'_k x_k + v'_k \end{aligned} \quad (7.29)$$

The covariance of the new measurement noise v' , and the cross covariance between the process noise w and the new measurement noise v' , can be obtained as

$$\begin{aligned} E[v'_k v'^T_k] &= E[(H_{k+1}w_k + \zeta_k)(w_k^T H_{k+1}^T + \zeta_k^T)] \\ &= H_{k+1}Q_k H_{k+1}^T + Q_k \zeta_k \\ E[w_k v'^T_k] &= E[w_k(w_k^T H_{k+1}^T + \zeta_k^T)] \\ &= Q_k H_{k+1}^T \end{aligned} \quad (7.30)$$

where we have used the fact that w_k and ζ_k are independent and zero-mean.

Now we will define the *a priori* and *a posteriori* state estimates for the system of Equation (7.29) slightly differently than we have up to this point. The state estimate $\hat{x}_k^-$ at time k is defined as the expected value of the state x_k conditioned on measurements up to and including time k .

$$\hat{x}_k^- = E[x_k | y_1, \dots, y_k] \quad (7.31)$$

The state estimate at time $\hat{x}_k^+$ at time k is defined as the expected value of the state x_k conditioned on measurements up to and including time $(k+1)$. We assume that it is given by a standard linear predictor/corrector combination:

$$\begin{aligned} \hat{x}_k^+ &= E[x_k | y_1, \dots, y_{k+1}] \\ &= \hat{x}_k^- + K_k(y'_k - H'_k \hat{x}_k^-) \end{aligned} \quad (7.32)$$

Note that these definitions of $\hat{x}_k^-$ and $\hat{x}_k^+$ are slightly different than the definitions used elsewhere in this book. Usually, $\hat{x}_k^-$ is based on measurements up to and including time $k-1$, and $\hat{x}_k^+$ is based on measurements up to and including time k . In this section, these two estimates are both based on one additional measurement. As in our previous derivations, we choose the gain K_k to minimize the trace of the covariance of the estimation error. In equation form this is written as

$$K_k = \operatorname{argmin} \operatorname{Tr} E[(x_k - \hat{x}_k^+)(x_k - \hat{x}_k^+)^T] \quad (7.33)$$

We will not work through the details here, but in [Bry68] it is shown that this minimization leads to the following estimator equations.

The discrete-time Kalman filter with colored measurement noise

1. Our system and measurement equations are given by Equation (7.26).
2. y'_k and H'_k are defined by Equations (7.27) and (7.28).
3. At each time step, execute the following equations to update the state estimate:

$$\begin{aligned}
\hat{x}_k^+ &= \hat{x}_k^- + K_k(y'_k - H'_k \hat{x}_k^-) \\
\hat{x}_{k+1}^- &= F_k \hat{x}_k^+ + C_k(y'_k - H'_k \hat{x}_k^+) \\
K_k &= P_k^- H_k'^T (H_k' P_k^- H_k'^T + R_k)^{-1} \\
M_k &= Q_k H_{k+1}^T \\
C_k &= M_k (H_k' P_k^- H_k'^T + R_k)^{-1} \\
P_k^+ &= (I - K_k H_k') P_k^- (I - K_k H_k')^T + K_k R_k K_k^T \\
P_{k+1}^- &= F_k P_k^+ F_k^T + Q_k - C_k M_k^T - F_k K_k M_k - M_k^T K_k^T F_k^T \quad (7.34)
\end{aligned}$$

A similar approach to the continuous-time filter with colored measurement noise is given in [Ste68].

■ EXAMPLE 7.2

Consider the following linear system with colored measurement noise:

$$\begin{aligned}
x_k &= \begin{bmatrix} 0.70 & -0.15 \\ 0.03 & 0.79 \end{bmatrix} x_{k-1} + \begin{bmatrix} 0.15 \\ 0.21 \end{bmatrix} w_{k-1} \\
y_k &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x_k + v_k \\
v_k &= \psi v_{k-1} + \zeta_{k-1} \\
E[w_k w_j^T] &= 1 \delta_{k-j} \\
E[\zeta_k \zeta_j^T] &= \begin{bmatrix} 0.05 & 0 \\ 0 & 0.05 \end{bmatrix} \delta_{k-j} \\
E[w_k \zeta_j^T] &= 0 \quad (7.35)
\end{aligned}$$

The scalar ψ indicates the correlation of the measurement noise. If $\psi = 0$ then the measurement noise is white. As ψ increases, the color of the measurement noise increases (i.e., it contains more low-frequency components and less high-frequency components). In this example, we simulate the Kalman filter for this system in three different ways. First, we simulate the standard Kalman filter while simply ignoring the colored nature of the measurement noise. Second, we augment the state vector as described in Section 7.2.2, which will take the colored nature of the measurement noise into account, and then simulate the Kalman filter. Third, we implement the measurement-differencing approach that is described in this section, which again takes the colored nature of the measurement noise into account, and then simulate the filter. Table 7.2 shows the experimental values of the trace of the covariance of the estimation error for the three filters. We can see that if $\psi = 0$ then the three filters perform essentially identically. (There is some difference in performance between the filters because the performance measures in Table 7.2 are experimentally determined statistical values.) However, as ψ increases (i.e., the color of the measurement noise increases) we see that the filters that take this into account provide increasingly better performance compared to the standard Kalman filter. This example shows the improvement in perfor-